表 7－1　正态总体均值、方差的置信区间与单侧置信限(置信水平为 $1-\alpha$)

	待估参数	其他参数	枢轴量的分布	置信区间	单侧置信限
单个正态总体	μ	σ^2 已知	$Z=\dfrac{\overline{X}-\mu}{\sigma/\sqrt{n}}\sim N(0,1)$	$\left(\overline{X}\pm\dfrac{\sigma}{\sqrt{n}}z_{\alpha/2}\right)$	$\overline{\mu}=\overline{X}+\dfrac{\sigma}{\sqrt{n}}z_\alpha$　$\underline{\mu}=\overline{X}-\dfrac{\sigma}{\sqrt{n}}z_\alpha$
	μ	σ^2 未知	$t=\dfrac{\overline{X}-\mu}{S/\sqrt{n}}\sim t(n-1)$	$\left(\overline{X}\pm\dfrac{S}{\sqrt{n}}t_{\alpha/2}(n-1)\right)$	$\overline{\mu}=\overline{X}+\dfrac{S}{\sqrt{n}}t_\alpha(n-1)$　$\underline{\mu}=\overline{X}-\dfrac{S}{\sqrt{n}}t_\alpha(n-1)$
	σ^2	μ 未知	$\chi^2=\dfrac{(n-1)S^2}{\sigma^2}\sim\chi^2(n-1)$	$\left(\dfrac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)},\dfrac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\right)$	$\overline{\sigma^2}=\dfrac{(n-1)S^2}{\chi^2_{1-\alpha}(n-1)}$　$\underline{\sigma^2}=\dfrac{(n-1)S^2}{\chi^2_\alpha(n-1)}$
两个正态总体	$\mu_1-\mu_2$	σ_1^2,σ_2^2 已知	$Z=\dfrac{(\overline{X}-\overline{Y})-(\mu_1-\mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}}\sim N(0,1)$	$\left(\overline{X}-\overline{Y}\pm z_{\alpha/2}\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}\right)$	$\overline{\mu_1-\mu_2}=\overline{X}-\overline{Y}+z_\alpha\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$ $\underline{\mu_1-\mu_2}=\overline{X}-\overline{Y}-z_\alpha\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$
	$\mu_1-\mu_2$	$\sigma_1^2=\sigma_2^2=\sigma^2$ 未知	$t=\dfrac{(\overline{X}-\overline{Y})-(\mu_1-\mu_2)}{S_W\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}}\sim t(n_1+n_2-2)$ $S_W^2=\dfrac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}$	$\left(\overline{X}-\overline{Y}\pm t_{\alpha/2}(n_1+n_2-2)\times S_W\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}\right)$	$\overline{\mu_1-\mu_2}=\overline{X}-\overline{Y}+t_\alpha(n_1+n_2-2)S_W\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$ $\underline{\mu_1-\mu_2}=\overline{X}-\overline{Y}-t_\alpha(n_1+n_2-2)S_W\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$
	$\dfrac{\sigma_1^2}{\sigma_2^2}$	μ_1,μ_2 未知	$F=\dfrac{S_1^2/S_2^2}{\sigma_1^2/\sigma_2^2}\sim F(n_1-1,n_2-1)$	$\left(\dfrac{S_1^2}{S_2^2}\dfrac{1}{F_{\alpha/2}(n_1-1,n_2-1)},\dfrac{S_1^2}{S_2^2}\dfrac{1}{F_{1-\alpha/2}(n_1-1,n_2-1)}\right)$	$\left(\overline{\dfrac{\sigma_1^2}{\sigma_2^2}}\right)=\dfrac{S_1^2}{S_2^2}\dfrac{1}{F_{1-\alpha}(n_1-1,n_2-1)}$ $\left(\underline{\dfrac{\sigma_1^2}{\sigma_2^2}}\right)=\dfrac{S_1^2}{S_2^2}\dfrac{1}{F_\alpha(n_1-1,n_2-1)}$